

Nmap Security Analysis Report

Report Date: Friday, October 2, 2026, 16:31:17 **Subject:** nmap_example.xml
Scan Date: January 27, 2008 **Scan Type:** SYN Stealth Scan (-sS) with
Service/Version Detection and OS Fingerprinting (-A)

1. Executive Summary

An analysis of the `nmap_example.xml` scan results reveals several critical security vulnerabilities and configuration weaknesses across the identified infrastructure. The primary concerns involve the use of deprecated cryptographic protocols, unauthorized access configurations, and excessive information disclosure through service banners.

2. Key Findings & Vulnerabilities

Critical Risk: Unauthorized Access & Data Exposure

- **Anonymous FTP Access:** The host `pub2.kernel.org` (204.152.191.37) is configured to allow anonymous FTP logins. This presents a high risk of unauthorized data exfiltration or the introduction of malicious files to the repository.

Medium Risk: Deprecated Cryptographic Protocols

- **SSLv2 Support:** Multiple high-traffic hosts, including `sourceforge.net`, `www.netbsd.org`, and Google Front End nodes (72.14.207.99, 72.14.253.83), were identified as supporting **SSLv2**. This protocol is cryptographically broken and susceptible to man-in-the-middle (MITM) attacks.

Low Risk: Information Disclosure

- **Service Version Leakage:** Numerous hosts (including `freshmeat.net`, 64.13.134.48, and `cvs.openbsd.org`) are broadcasting specific software version numbers (e.g., Apache 1.3.39, PHP 4.4.7, OpenSSH 4.3). This allows attackers to perform targeted reconnaissance and identify specific exploits for known vulnerabilities in those versions.
- **Service Exposure:** Several hosts are exposing management and file-transfer protocols (SSH, FTP, rsync, and Finger) directly to the public internet, increasing the attack surface.

3. Infrastructure Overview

The scan identified a diverse range of operating systems, primarily centered around Linux (various kernels) and BSD-based systems (NetBSD, OpenBSD).

The infrastructure includes significant web, DNS, and file distribution services (rsync, AFS) essential for large-scale software repositories.

4. Recommendations

1. **Disable SSLv2:** Immediately disable support for SSLv2 and SSLv3 across all web-facing services, transitioning to TLS 1.2 or higher.
2. **Restrict FTP Access:** Reconfigure `pub2.kernel.org` to disallow anonymous FTP logins unless strictly required for public distribution, and implement strict directory permissions.
3. **Harden Service Banners:** Configure web servers (Apache, lighttpd) and SSH services to suppress version strings in HTTP headers and service banners to mitigate reconnaissance efforts.
4. **Audit Management Ports:** Review the necessity of exposing ports such as 21 (FTP), 22 (SSH), and 79 (Finger) to the public internet; implement IP whitelisting or VPN requirements where possible.